Innovative Pathways to Data Literacy: Tailored Formats for Humanities and Cultural Studies

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In today's research landscape, data competencies are indispensable for tackling complex interdisciplinary questions. Yet, scholars in the humanities and cultural studies often face challenges in digitising analogue materials, analysing datasets with digital methods, and integrating qualitative approaches with data-driven techniques. This highlights the need for tailored training formats that address diverse levels of expertise with digital tools, prioritising researchers' specific needs over their career stages.

To address this need, the HERMES Data Competence Centr e (Humanities Education in Research, Data, and Methods), funded by the German Federal Ministry of Education and Research (BMBF) 2023-2026, works on the implementation of innovative formats in the humanities and cultural studies to enrich curricula by integrating digital methods, boosting digital literacy, and advancing methodological competencies. Among these, Bring Your Own Data Lab (BYODL) and Data Carpentries stand out as complementary approaches to enhancing data literacy. Although such formats are more widely established in fields like the natural sciences[1] , they remain underutilised in the humanities and cultural studies, presenting untapped potential for adaptation to these fields while customising to their specific needs. By aligning both formats, HERMES bridges the gap between traditional scholarship and the demands of an increasingly data-driven academic landscape. By organizing free interdisciplinary workshops both online and on-site, HERMES creates the opportunity for further targeted Data Literacy under real-time guidance of experts. It offers a structured yet flexible learning pathway that empowers researchers with varying levels of experience in digital methodologies to acquire, refine, and apply data skills in practical, inclusive, and impactful ways.

Data Carpentries , part of the non-profit organisation The Carpentries , provides domain-specific lessons to "teach fundamental data skills needed to conduct research" (The Carpentries, 2024b), thereby laying the foundation for individual learning paths. Collaboratively developed and maintained by a global, interdisciplinary community based on a well-tested pedagogic concept, these open-source lessons are delivered by certified instructors in workshops. The content is grounded in real-world scientific questions and datasets and ensures practical relevance (Teal et al., 2015). Although The Carpentries' format is widely adopted in the natural sciences, effectively addressing learners' needs (Crall, Barnes, and Word, 2024), it has no equivalent curriculum for the humanities yet (Barnes, Michonneau, and Jordan, 2021). To bridge this gap, we create lessons tailored to humanities researchers new to data-driven research, delivered in free workshops to refine the materials and promote both the content and the hands-on Carpentries teaching approach within the humanities community.

Building on this foundational knowledge, **BYODL** offers researchers a platform to advance their skills and apply digital methods directly to their datasets. While participation in Data Carpentries workshops is not mandatory, BYODL participants are expected to have essential data-handling skills. The format emphasizes collaboration and practical application, enabling researchers to process, analyse, and visualise their data while addressing specific research questions. Our adaptation of BYODL for the humanities and cultural studies equips early-career researchers, scholars, and GLAM staff with the tools to navigate sophisticated digital workflows and enhance their methodological expertise. Guided by experts, participants gain a deeper understanding of their datasets by addressing their individual inquiries, enabling more effective application across diverse scholarly contexts. The collaborative and stimulating environment brings together participants from various disciplines, fostering valuable exch-

anges and insights into different approaches to data and methodology.

Together, these formats create a structured pathway to data literacy: **Data Carpentries** lays the groundwork for foundational skills, while **BYODL** provides advanced, applied learning opportunities. This integration bridges traditional studies with digital methods, demystifies digital tools, fosters interdisciplinary collaboration, and supports Open Science principles by promoting transparency, inclusivity, and equitable access to data-driven research practices. Our poster highlights the complementary strengths of these formats, showcasing their implementation within HERMES to create a robust, and sustainable approach to data literacy in humanities and cultural disciplines. Application examples illustrate the impact on participants' research and demonstrate how models, adapted from other disciplines, can be effectively tailored to these disciplines to address existing gaps. Finally, we explore how these formats foster practical, peer-to-peer learning environments that support data-driven skill development. Through this poster, we aim to inspire dialogue on scaling and adapting similar initiatives to further integrate digital methods into curricula and build a more inclusive, data-literate academic community.

Notes

1. Some examples for BYODLs can be the workshops from the *Dutch Techcenter for Life Sciences* and *elixir* (Bernabé et al. (2023). A more general approach is conducted by the "Bring Your Own Data Working Groups" organised by the Northwestern University (2024). In the case of Data Carpentries, curricula are available for life sciences as well as astronomy and social sciences (The Carpentries 2024a).

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